Kinetics w/ DFT

1) Discuss TST and calculating k's from DFT data

2) Review of basic principles of microkinetic modelling

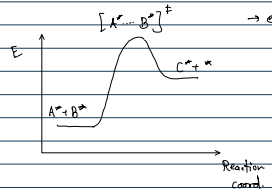
3) Scaling relationships, BEP reations, volcano plots, trends in catalytic properties

Transition State Theory (TST)

Strategy for estimating rate constants from DFT (exp.) results

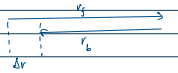
Elementary surface reaction

$A^* + B^* \rightleftharpoons C^* + *$



Two key assumptions:

1) To a first order perturbation, reactants are in (quasi) equilibrium w/ T.S.



↳ or else there would be no rate (if exactly at equilibrium)

2) Rate of conversion of TS to products governed by attempt frequency, ν

↳ (vibrational frequency that dictates how it "vibrates" from

(i) $K^\ddagger = e^{\frac{-\Delta\mu^{o\ddagger}}{k_B T}} = \frac{a_{TS}}{a_{A^*} a_{B^*}}$

$K^\ddagger$ → standard state chemical potential for transition state

$\Delta\mu^{o\ddagger} = \mu^o_{TS} - \mu^o_{A^*} - \mu^o_{B^*}$

Recall: $\mu_i = \mu_i^o + k_B T \ln a_i$; $a_i = \frac{f_i}{f_i^o}$; μ_i^o depends on T

(a) For ideal gases, $a_i = \frac{P_i}{P^o}$

(b) For liquid species,

$a_i \approx$ { 1 if conc. (usually solvent); $\frac{c_A}{c_A^o}$ if dilute "i"; c_A^o ← infinitely dilute solute extrapolated to 1M (of Henry's law) }

(c) Surface species: Use partition functions and divide into purely local part and a configurational part

Note: $a_{A^*} \approx \theta_A$

(This usually gives right answer but is not that rigorous)

Assume: non-interacting adsorbates on lattice N_s sites. Adsorption energy is ε for each adsorbate. Each adsorbate has local energy characterized by q_{loc} (vibration, rotation)

$Q_{tot}(T, N_s, N) = \frac{N_s!}{(N_s - N)!\,N!} \left(e^{-\beta\varepsilon} q_{loc}\right)^N$ ↳ canonical

$F = -k_B T \ln Q_{tot}$ extensive

$\ln Q_{tot} = -\beta\varepsilon N + N \ln q_{loc} + \ln N_s! - \ln(N_s - N)! - \ln N!$; $\ln x! = x \ln x - x$

$= N_s \ln\left(\frac{N_s}{N_s - N}\right) + N \ln\left(\frac{N_s - N}{N}\right) - \beta\varepsilon N + N \ln q_{loc}$

$F = -k_B T \ln Q$, $\mu = \left.\frac{\partial F}{\partial N}\right|_{T, N_s} = \underbrace{\varepsilon - k_B T \ln q_{loc}}_{\text{adsorption free energy for an isolated molecule}} + \underbrace{k_B T \ln \frac{\theta}{1-\theta}}_{\text{differential configurational entropy}}$; $\theta = \frac{N}{N_s}$

} only approximation is that they are non-interacting

If more than one adsorbate is present, $\mu_i = \varepsilon_i - k_B T \ln q_{trc,i} + k_B T \ln \frac{\theta_i}{\theta_*}$; $\theta_* = 1 - \theta_i - \theta_B - \theta_C - \dots$

We say that $\mu_i^\circ = \varepsilon_i - k_B T \ln q_{trc,i}$; $a_i = \frac{\theta_i}{\theta_*}$

Standard state : isolated molecule w/ no configurational entropy

↳ For reaction: $A^* + B \overset{TS}{\rightleftharpoons} C^* + *$, $K^\ddagger = \frac{q_{TS}}{q_{A^*} q_{B^*}} = \frac{\theta_{TS}}{\theta_A \theta_B}$; $\ln K^\ddagger = \frac{\mu_{TS}^\circ - \mu_{A^*}^\circ - \mu_{B^*}^\circ}{k_B T}$

Second assumption: $r \approx \nu \theta_{TS} \theta_*$ (assumes immediate TS (Langmuir Hinshelwood))

$\therefore r = \nu K^\ddagger \theta_A \theta_B = \underbrace{\nu e^{\frac{\Delta\mu^\ddagger}{k_B T}}}_{\text{what about this?}} \theta_A \theta_B$ (Law of mass action)

For particles on surface, $\mu_i^\circ = \varepsilon_i - k_B T \ln q_{trc}$

$= -k_B T \ln e^{-\beta \varepsilon_i} - k_B T \ln q_{trc}$

$= -k_B T \ln \left[q_{trc,i} e^{-\frac{\varepsilon_i}{k_B T}} \right]$

q_i (includes local kinetic & potential energies)

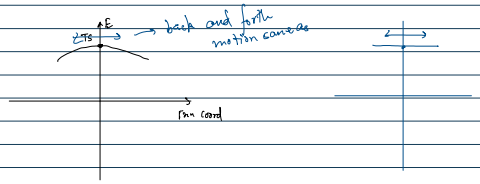
$\therefore \mu_i^\circ = -k_B T \ln q_i$

$\Delta\mu^\ddagger = \mu_{TS}^\circ - \mu_{A^*}^\circ - \mu_{B^*}^\circ = -k_B T \ln \left(\frac{q_{TS}}{q_{A^*} q_{B^*}} \right)$ ⇒ $r \approx \nu \frac{q^\ddagger}{q_{A^*} q_{B^*}} \theta_A \theta_B$; $q^\ddagger = q_{TS}$

To find ν: write q_i's as : $q_i = \sum e^{-\beta E_{vj,i}} = e^{-\beta E_{DFT,i}} \cdot q_{vib,i}$ (like ε; Assume only harmonic (parabolic energy potential) vib modes)

$= e^{-\beta E_{DFT,i}} e^{-\beta (ZPE)_i} \prod_{j=1}^{3N} \left(\frac{1}{1 - e^{-h\nu_j / k_B T}} \right)$

For mode "j" along reaction coord ($A^* \cdots B^*$)



Generally assumed we can treat this as a curvature of zero, but attempt frequency is accurate

$\frac{1}{\left(1 - e^{-h\nu / k_B T}\right)} \approx \frac{k_B T}{h \nu_j}$

$\therefore q^\ddagger = e^{-\beta E_{TS}^{DFT}} q_{vib}^{\nu} \cdot \frac{k_B T}{h \nu_j}$

q_{vib}^{ν}: all modes except rxn coord.

$r \approx \frac{\nu \frac{k_B T}{h \nu_j} e^{-\beta E_{TS,DFT}} q_{vib,TS}^{\nu}}{e^{-\beta E_{A^*DFT}} e^{-\beta E_{B^*DFT}} q_{vib,A^*} q_{vib,B^*}}$; If $\nu = \nu_j$,

$r = \frac{k_B T}{h} \frac{q_{vib,TS}^{\times}}{q_{A^*,vib} q_{B^*,vib}} e^{\beta \Delta E_a^{ZPE}} \theta_A \theta_B$

(× does not include ZPE; includes potential E & ΔZPE)

This can be written as $r = \nu e^{-\frac{\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B$

$= \nu \frac{k_B T}{h \nu} e^{-\frac{\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B$

$r = \frac{k_B T}{h} e^{-\frac{\Delta\mu^\ddagger}{k_B T}} \theta_A \theta_B$

(k (rate constant))

3) Rate is proportional to the coverage of TS on substrate

Transition State Theory w/ gaseous species

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k (rate constant)

Transition State Theory w/ gaseous species

$A(g) + B^* \xrightarrow{TS^*} C^*$ (treat TS as a surface species which is often a good assumption)

$$K^\ddagger = \frac{a_{TS}^*}{a_A a_B^*} = \frac{\theta_{TS}/\theta_*}{\frac{p_A}{p^\circ}\frac{\theta_B}{\theta_*}} \Rightarrow \theta_{TS} = K^\ddagger \theta_B \frac{p_A}{p^\circ}$$

Thus, $r = \nu \theta_{TS} = \nu K^\ddagger \theta_B \frac{p_A}{p^\circ} = \frac{k_B T}{h} K^\ddagger \theta_B \frac{p_A}{p^\circ}$

Now including rxn coord.